

How Competitive is Modern Sumo Wrestling?

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1 Introduction

In this paper, we assess the competitive balance of professional sumo wrestling over a period of nearly 25 years. Sumo wrestling has many features that make it unique compared to other sports. It is an individual sport¹ where a large pool of athletes face off against their competitors many times per year at a variety of locations. Specifically, there are six tournaments per year, and in the top two professional divisions (which we examine in this paper), each wrestler has 15 bouts per tournament. Sumo wrestling has a total of six divisions, with the top two divisions (Makuuchi and Juryo) being considered the serious professional levels. After each tournament, wrestlers are promoted or demoted to different ranks (within or across divisions) based on their performance, with additional champion ranks being available to the highest performing athletes. Further, professional sumo wrestling has no weight or height categories (unlike many other combat sports), which can lead to significant mismatches in size and strength between competitors. Another unique feature of sumo wrestling is the short “game time” in each bout, which may only last a few seconds. A fight is considered over when one wrestler exits the ring or touches the ground with any part of their body other than the soles of their feet. Since wrestlers can feint or dodge their opponents and due

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¹While sumo wrestlers live and train together in groups called heya or stables, fans tend to root for individual wrestlers rather than an entire stable.

to the very short duration of the fight, it may seem that randomness is a significant factor in determining the winner of a sumo bout.

Since each wrestler fights 90 times per year against a wide variety of opponents, there are many games and matchups to analyze. Fortunately, a public database posts the results of every professional sumo bout, along with the ranks, heights, weights, etc. of each wrestler in advance of every tournament. This gives us a rich set of data to analyze over a long period of time. We scrape the public database to amass a database of over 70,000 bouts spanning from 2000 to 2025. Looking at this long time period allows us to assess the competitive balance of sumo wrestling over time, as well as to identify the overall predictability of professional sumo.

1.1 Data Preparation

The source of our sumo bouts is a public repository of sumo data, <http://sumodb.sumogames.de/>. The site has two databases, one at the wrestle level and one at the bout level. Both databases may only be queried 1000 rows at a time, so we wrote webscraping scripts to programatically download all bouts involving a Makuuchi or Juryo (top two divisions) wrestler from 2000 to 2025.² We then scraped the player database to obtain the demographic information for all wrestlers in the bout data.

From the bouts data, we extracted the following variables: tournament date, day of tournament³, wrestler name, rank, winning technique, and wrestler name and rank for the opponent. From the player data, we extracted: wrestler name, stable, hometown, birth date, sumo debut date, height (in cm), weight (in kg), current rank, age, and record. Interestingly, the player database is updated every tournament, and so we can observe changes in within-wrestler weight over time. We merged the datasets on wrestler name, date, and rank to

²There is a September tournament ongoing at the time of writing, which we exclude due to incomplete tournament results. Additionally, because some fights occasionally occur between two wrestlers from different divisions, we include a handful of Makushita (third division) wrestlers who fought a Juryo opponent.

³tournaments typically last 15 days.

create a bout-level dataset with $[[X]]$ rows.⁴

Once downloaded and combined, we created additional variables to use as features in our analysis. We subtract tournament date by sumo debut date to get years of experience. From the player rank, we create a numerical rank variable that accounts for the hierarchy of sumo divisions.⁵ Next, we create a series of variables to capture differences in attributes between bout opponents, including age difference, weight difference, height difference, experience difference, and ranking difference. We create a series of indicator variables that may capture other effects, including whether the bout is on the first or last day of the tournament, whether the bout is between wrestlers of different divisions, and whether the bout is between two wrestlers from the same hometown.

Taking advantage of the panel nature of our data, we create a series of variables to capture historical performance. For each wrestler, we calculate both their all-time cumulative win-rate and their win-rate over the previous six tournaments (12 months). Calculating historical performance allows us to control for a wrestler’s opponent quality and to identify the difference in historical performance between two wrestlers in a bout.

Lastly, in addition to our bout-level dataset, we create a tournament-level dataset by aggregating the bout-level data to the tournament-wrestler level. In aggregating the data, we calculate the mean of each demographic variable for the opponents of each wrestler in each tournament. We filter this dataset to exclude wrestlers with fewer than 7 bouts in a tournament as a proxy for being injured. This leaves us with $[[X]]$ tournament-wrestler observations.

⁴This represents $[[X]]$ unique bouts, where each bout is recorded twice, once for each wrestler. We create a “winner” variable to indicate which wrestler won the bout.

⁵While a full discussion of the sumo ranking system is outside the scope of this paper, each division uses a field ranking system in which, e.g. rank 4 is considered better than rank 10. Because Makuuchi is a strictly better division than Juryo, our numerical ranking would reflect a Makuuchi rank 10 as being better than a Juryo rank 2, for example.

2 Methods

There are many possible ways to analyze competitiveness in a sport. Perhaps the simplest way is to look at the standard deviation of win rate by competitor. We do this with our tournament-level dataset, where the point of interest is in analyzing the standard deviation of win rate over time. If sumo wrestling has become more competitive over time (perhaps due to more widespread adoption of training techniques, nutrition, etc.), we would expect to see the standard deviation of win rate decrease over time. Conversely, if sumo wrestling has become less competitive over time (perhaps due to a concentration of dominant wrestlers), we would expect to see the standard deviation of win rate increase over time. Further, instead of looking at the standard deviation of win rate, we can graph the distribution of win rates over time to look for more nuanced changes.

An alternative way of assessing the competitiveness of sumo is to look at the predictability of individual bouts. If being taller or heavier than your opponent provides a massive winning advantage, then sumo may not be competitive and the conclusion is that heavier opponents will win much more. Conversely, if the outcome of a bout is very difficult to predict based on observables, then this provides evidence that sumo is a competitive sport.⁶ We use two approaches to modelling bout-level outcomes. First is a logistic regression model, which models the log odds of winning as a linear function of each control variable. Second is a LightGBM model, which is a gradient-boosted decision tree classifier that can identify more complex and nuanced relationships among the controls. In each model, we are concerned with the model fit as a measure of how well the observables can explain the outcome of a bout.

⁶We must clarify that at least from a spectator viewpoint, there is a difference between competitiveness and randomness. While professional coin flipping is mathematically a “competitive” sport, fans probably think of competitiveness as equal probability of outcome due to skill. In the case of sumo, we believe that skill plays at least a significant part in determining the outcome of a bout, and so our analysis is not just capturing randomness.

2.1 Modeling

3 Results and Analysis

3.1 Summary Analysis

3.2 Correlation Analysis

3.3 Models

3.4 Model Predictions

All results in this section use the LightGBM model with the highest R^2 from the table above, unless otherwise specified.

3.4.1 Sentiment Contribution

4 Discussion

5 Conclusion

6 Appendix